A. United We Stand

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

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https://codeforces.com/problemset/problem/1859/A

Given an array a of length n, containing integers. And there are two initially empty arrays b and c. You need to add each element of array a to **exactly one** of the arrays b or c, in order to satisfy the following conditions:

- Both arrays b and c are non-empty. More formally, let l_b be the length of array b, and l_c be the length of array c. Then $l_b, l_c \geq 1$.
- For any two indices i and j ($1 \leq i \leq l_b$, $1 \leq j \leq l_c$), c_j **is not** a divisor of b_i.

Output the arrays b and c that can be obtained, or output -1 if they do not exist.

Input

Each test consists of multiple test cases. The first line contains a single integer t ($1 \leq t \leq 500$) — the number of test cases. The description of the test cases follows.

The first line of each test case contains a single integer n ($2 \leq n \leq 100$) — the length of array a.

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($1 \leq a_i \leq 10^9$) — the elements of array a.

Output

For each test case, output a single integer -1 if a solution does not exist.

Otherwise, in the first line, output two integers l_b and l_c — the lengths of arrays b and c respectively.

In the second line, output l_b integers $b_1, b_2, \ldots, b_{l_b}$ — the elements of array b.

In the third line, output l_c integers $c_1, c_2, \ldots, c_{l_c}$ — the elements of array c.

If there are multiple solutions, output any of them. You can output the elements of the arrays in any order.

Example

Input

```
5
3
2 2 2
5
1 2 3 4 5
3
1 3 5
7
1 7 7 2 9 1 4
5
4 8 12 12 4
```

Output

```
-1
3 2
1 3 5
2 4
1 2
1
3 5
2 5
1 1
2 4 7 7 9
3 2
4 8 4
12 12
```

Note

In the first test case, a solution does not exist.

In the second test case, we can obtain b = [1, 3, 5] and c = [2, 4]. Then elements 2 and 4 do not divide elements 1, 3 and 5.

In the fifth test case, we can obtain b = [4, 8, 4] and c = [12, 12].